

LIBRARY MANAGEMENT SYSTEM WITH QR CODE-BASED BOOK TRACKING

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ABSTRACT

This study develops a web-based library management system with QR code-based book tracking developed in Django, reducing average book retrieval time by 52% and inventory discrepancy by 78%.

INTRODUCTION

The CvSU Imus Campus library relies on manual card-based borrowing records, resulting in frequent inventory discrepancies and slow book retrieval. A digitized library system with automated tracking can significantly improve efficiency. This study designs and implements a Django-based Library Management System (LMS) with QR code-based book identification for the campus library's 8,500-volume collection.

METHODOLOGY

The system was built using Django 4.2 (Python) with PostgreSQL as the database backend and Bootstrap 5 for the frontend. QR codes are generated using the qrcode library (one code per book copy). A Raspberry Pi 3 with a USB QR scanner serves as the circulation desk terminal. Modules include: catalog management, borrowing/return workflow, overdue tracking, and inventory audit. User acceptance testing was conducted with 5 librarians and 50 students over 4 weeks.

RESULTS

Average book retrieval time decreased from 4.2 minutes to 2.0 minutes (52% reduction). Inventory discrepancy rate dropped from 14% to 3.1% (78% improvement). Overdue detection accuracy was 100% compared to 73% for the manual system. System response time averaged 1.1 seconds. All 5 librarians rated system usability as 'Excellent' on the SUS scale (avg score: 87.4).

DISCUSSION

QR-based tracking combined with a digital workflow substantially improves library operations. The 3.1% remaining inventory discrepancy is attributed to unreturned books from before system adoption. Future enhancements should include RFID for hands-free scanning, integration with the university's student information system, and a student-facing mobile app

for catalog browsing.